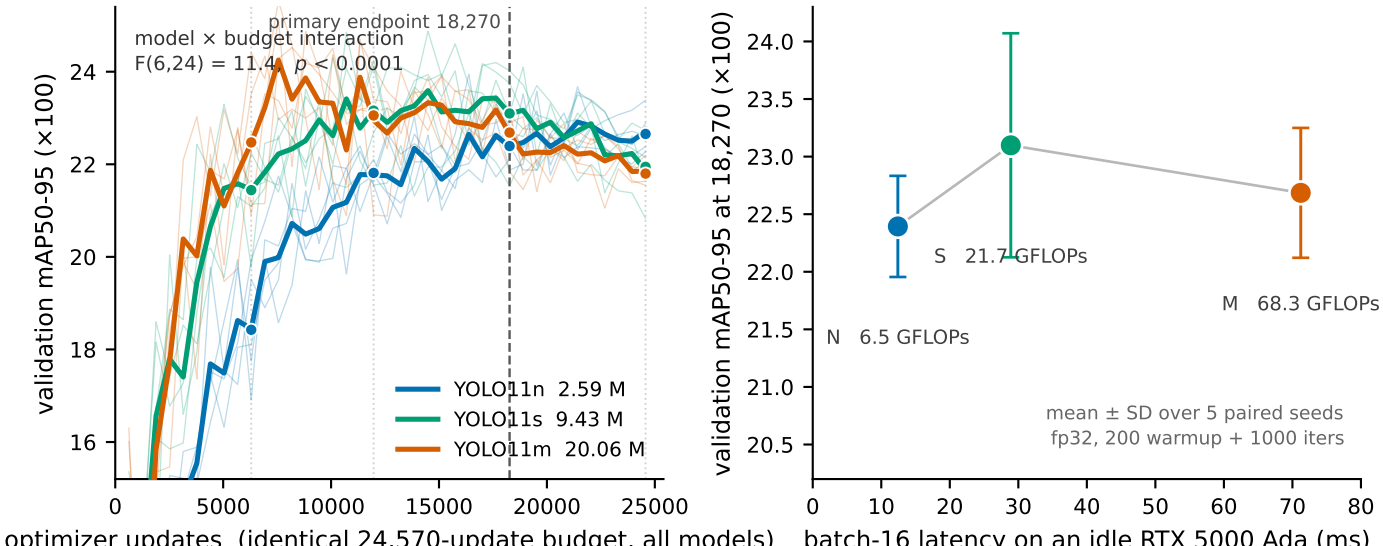
(a) Capacity helps early, then reverses
primary endpoint 18,270
model × budget interaction
F(6,24) = 11.4, p < 0.0001
validation mAP50-95 (×100)
YOLO11n 2.59 M
YOLO11s 9.43 M
YOLO11m 20.06 M
optimizer updates (identical 24,570-update budget, all models)
(b) 5.7× the latency buys nothing
validation mAP50-95 at 18,270 (×100)
N 6.5 GFLOPs
S 21.7 GFLOPs
M 68.3 GFLOPs
mean ± SD over 5 paired seeds
fp32, 200 warmup + 1000 iters
batch-16 latency on an idle RTX 5000 Ada (ms)